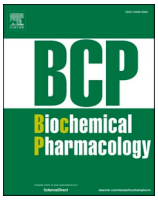
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Knockdown of Galectin-9 alleviates rheumatoid arthritis through suppressing TNF-α-induced activation of fibroblast-like synoviocytes

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ABSTRACT

The role of Galectin-9 (Gal-9) in the pathogenesis of rheumatoid arthritis (RA) remains unclear. This study aimed to investigate the mechanism of action and therapeutic potential of Gal-9 in RA. We detected Gal-9 expression in clinical samples, explored the mechanism of function of Gal-9 by knockdown and overexpression in fibroblast-like synoviocytes (FLSs), and further verified it in collagen-induced arthritis (CIA) model. We found that the levels of Gal-9 were considerably elevated in RA synovium than in osteoarthritis (OA) patients. A substantial decrease of Gal-9 was demonstrated after tumor necrosis factor (TNF-α) inhibitor treatment in the plasma of patients with RA. Additionally, transcriptome sequencing revealed that Gal-9 was involved in the regulation of the TNF-α pathway. Gal-9 was considerably upregulated after TNF-α stimulation in FLSs, and knockdown of Gal-9 substantially inhibited TNF-α activated proliferation, migration and inflammatory response. According to cell transcriptome sequencing results, we further confirmed that Gal-9 could achieve these effects by interacting with MAFB and affecting PI3K/AKT/mTOR pathway. Finally, we knocked down Gal-9 on the CIA model and found that it could alleviate the progression of arthritis. In conclusion, our study revealed that the knockdown of Gal-9 could inhibited TNF-α induced activation in RA through MAFB, PI3K/AKT/mTOR.

Abbreviations: **AAV**, adeno-associated virus; **ACPA**, anticitrullinated protein antibody; **ACR**, American College of Rheumatology; **AKT**, alpha serine/threonine-protein kinase; **CCK-8**, cell proliferation assay; **CIA**, collagen-induced arthritis; **CCL2**, chemokine ligand 2; **CO-IP**, Co-immunoprecipitation assay; CRP, C-reactive protein; **csDMARDs**, conventional synthetic disease-modifying anti-rheumatic drugs; **DAS28-ESR**, disease activity score in 28 joints for erythrocyte sedimentation rate; **DEGs**, Differentially Expressed Genes; **ESR**, erythrocyte sedimentation rate; **ELLISA**, enzyme-linked immunosorbent assay; **FLSs**, fibroblast-like synoviocytes; **Gal-9**, Galectin-9; **GO analysis**, Gene Ontology analysis; **HAQ**, health assessment questionnaire; **HE**, Haematoxylin and eosin; **IF**, immunofluorescence; **IHC**, Immunohistochemistry; **IL-1β**, interleukin-1β; **IL-6**, interleukin-6; **IL-8**, interleukin-8; **MAFB**, musculoaponeurotic fibrosarcoma oncogene homolog B; **MMP1**, matrix metalloproteinases 1; **MMP2**, matrix metalloproteinases 2; **MMP3**, matrix metalloproteinases 3; **MMP13**, matrix metalloproteinases 13; **mTOR**, Mammalian target of rapamycin; **NSAIDs**, non-steroidal anti-inflammatory drugs; **OA**, osteoarthritis; **PBMCs**, plasma and peripheral blood mononuclear cells; **PI3K**, Phosphoinositide 3-kinase; **PGA**, patient global assessment; **PHGA**, physician global assessment; **siRNA**, small interfering RNA; **SW28**, swelling joint 28 count; **RA**, rheumatoid arthritis; **RF**, rheumatoid factor; **TNF-α**, tumor necrosis factor-α; **TNFi**, tumor necrosis factor inhibitor; **T28**, tenderness joint 28 count; **WB**, western blot.

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1. Introduction

Rheumatoid arthritis (RA) is a common autoimmune disease characterised by synovial hyperplasia, inflammatory cell infiltration, and invasive pannus, which are triggered in a vicious cycle leading to synovitis and symmetrical destructive arthropathy [1]. The core of the lesion is an unrestricted "cancer-like" proliferation of synovial fibroblast-like synoviocytes (FLSs), which leads to multiple joint destruction, osteoporosis, systemic inflammation, and involvement of other important organs, resulting in a serious decline in quality of life and shortened life expectancy [2]. Patients with RA with longer disease duration have more organ damage and worse outcomes [3]. Tumor necrosis factor (TNF) plays an important role in RA as a pro-inflammatory factor by regulating the proliferation and differentiation of lymphocytes, promoting the proliferation of synoviocytes, secretion of metalloproteinases and cytokines, and interfering with the action of regulatory T cells [4].

Macrophage-derived TNF-α is one of the important factors in the early activation of FLSs in RA [5]. During the development of RA, after FLSs activation, fibroblast-like synoviocytes (FLSs) form the lining structure of the joints, proliferated abnormally, and were impaired by apoptosis [6]. Activated FLSs became invasive, leading to destruction of cartilage and bone, and FLSs also digested cartilage through the production of interleukins and the secretion of matrix metalloproteinases [2,7].

Gal-9, a key immunomodulatory factor, was involved in immunomodulation of tumors, viral infections, and autoimmune diseases [8,9]. The interaction between Gal-9 and Tim-3 played a key role in tumor immunity, and blocking the Gal-9/Tim-3 pathway induced anti-tumor immune responses and inhibited tumor growth [10,11]. A recent study identified the ERMAP-gal-9-dectin-2 axis as an "eat-me" signal for Kupffer cells that inhibited metastasis of hepatocellular carcinoma cells and exerted anti-tumor immune effects[9]. Plasma Gal-9 is considerably elevated in COVID-19 patients and positively correlated with a variety of pro-inflammatory cytokines and chemokines, suggesting that inhibition of Gal-9 may be used as a therapeutic approach for the treatment of COVID-19 infection [12]. In immune checkpoint inhibitor-emergent gouty arthritis, arthritis with crystal deposition locally exhibited down-regulation of TIM-3 expression on the surface of T cells, and disruption of the TIM-3/Gal-9 axis was a potential mechanism for T cell activation in gouty arthritis [13]. High levels of Gal-9 were associated with tumor progression in tumor-associated dermatomyositis [14]. Our previous study found that plasma Gal-9 expression was higher in patients with RA than in healthy controls and positively correlated with disease activity [15], Tacrolimus treatment reduced plasma GaL-9 levels and Gal-9 may serve as a biomarker for evaluating RA disease activity and treatment efficacy [16]. Additionally, exogenous Gal-9 inhibited the value-added and promoted apoptosis of RA-FLSs compared to Gal-1, 3, and 8 in a concentration-dependent manner [17]. However, endogenous Gal-9 presented an inconsistent effect with exogenous Gal-9, siRNA knockdown of Gal9 induced apoptosis in RA synovial fibroblasts, suggesting that endogenous Gal9 protected against apoptosis rather than inducing apoptosis compared to exogenous Gal9 [18].These results suggested that Gal-9 played an important role in RA-FLSs apoptosis and value-addition, however, the specific mechanism of its role and its relationship with FLSs activation were not clear.

In recent studies, the use of neutralizing antibodies to Gal-9 has often been chosen in cases where Gal-9 inhibition is required [19,20]. Gene therapy has been recognized as a potential approach for targeted therapy [21], such as gene therapy using viral vectors [22,23]. Due to the selectivity and safety of adeno-associated virus (AAV), we chose AAV as a vector to study the function of various molecules in animal models [24]. However, whether AAV-Gal-9 can be successfully constructed and affect inflammatory arthritis is unknown.

In this study, we stimulated RA-FLSs with TNF-α to mimic the inflammatory microenvironment of synoviocytes, and used CIA model to study the critical role of Gal-9 and explored the underlying mechanisms. Our results suggest that Gal-9 play an important role in alleviating TNF-α-induced inflammatory responses in RA-FLSs by binding to MAFB or through the PI3K-AKT pathway.

2. Materials and methods

2.1. Patient cohort and samples

The investigations were carried out following the rules of the Declaration of Helsinki of 1975, revised in 2013. All patients signed an informed consent form, and patients with RA met the diagnostic criteria of the American College of Rheumatology/European League Against Rheumatism 2010. Patients whose disease activity score in 28 joints for erythrocyte sedimentation rate (DAS28-ESR) $>$ 3.2 were enrolled in Department of Rheumatology and Immunology, Linyi People's Hospital from October 2021 to October 2022 (Approval no. YX200237). Patients meeting the criteria (shown in the next paragraph) received 24 weeks of treatment with the TNF-α Monoclonal Antibody Adalimumab (40 mg) every 2 weeks. Clinical assessment of arthritis and peripheral blood samples were collected at baseline and at 12 and 24 weeks, In the end, we obtained clinical data before and after treatment for 10 patients, peripheral blood serum specimens for 7 patients, and only 4 pairs of RNA samples remained after RNA sequencing.

The inclusion criteria were: patients with RA aged $>$ 18 years who received MTX for at least 12 consecutive weeks and had been on a stable dose of the study drug for $\geq$ 4 weeks ($\geq$10 mg/week); had been off csDMARDs (except MTX) for $\geq$ 4 weeks; non-steroidal anti-inflammatory drugs (NSAIDs), discontinued for at least 1 week, and a glucocorticoid dose $\leq$ 10 mg/day at a stable dose for at least 4 weeks. The exclusion criteria were: patients previously treated with Adalimumab or other biological DMARDs; patients with severe infections, hepatitis, tuberculosis, tumors, or with extra-articular organs involvement.

Synovial tissue samples were collected from surgical specimens of knee arthroplasty in patients clinically diagnosed with RA and OA (3 RA and 3 OA, Sex, age, disease activity matched) at Qilu Hospital of Shandong University between April 2022 and November 2022 (Approval no. KYLL-202000–165-1).

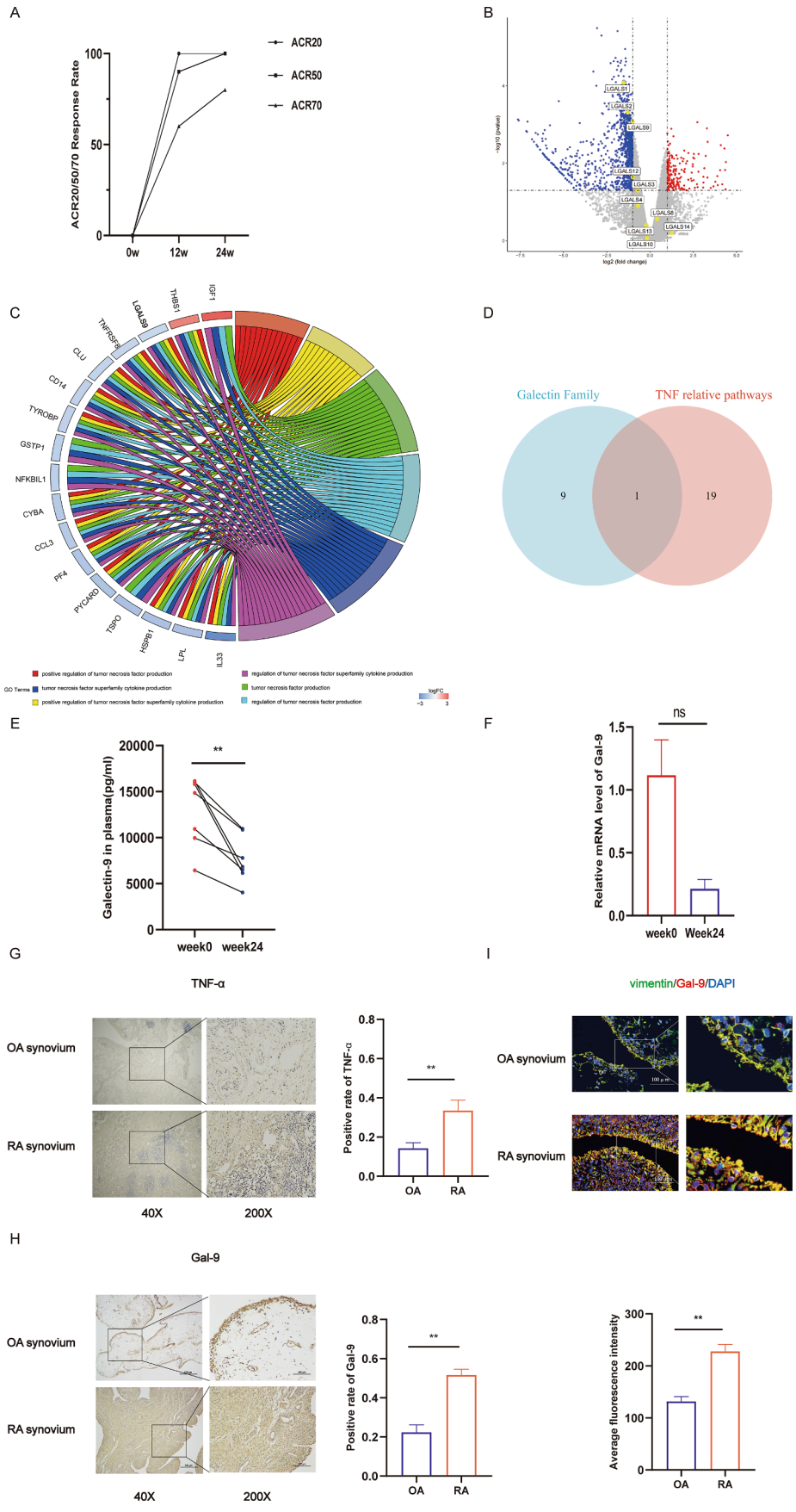
2.2. Cell isolation and culture

Human RA-FLSs (CTCC-001–0385) were purchased from Zhejiang Missen Cell Technology Co., Ltd. Three RA fresh synovial tissues and three OA fresh synovial tissues were minced and digested with type I collagenase (Sigma-Aldrich, USA) and then shaken for 3–4 h at 37 °C (100–120 r/min) on a shaker. Purchased and self-extracted cells were cultured in DMEM containing 10 % fetal calf serum, 100 IU/mL penicillin, and 100 μg/mL streptomycin in an incubator at 37 °C under 5 %

Table 1
The characteristic of 10 RA patients with Adalimumab treatment.

	Patients with RA under Adalimumab treatment (n = 10)		
	Week0	Week12	Week24
Male	3 (30 %)	–	–
Age, year	45.60 ± 3.26	–	–
Duration, month	23.0 (10.5,24.0)	–	–
RF positive	8 (80 %)	–	–
ACPA positive	10 (100 %)	–	–
ESR (mm/h)	51.50 ± 6.82	25.70 ± 6.25*	19.70 ± 5.48*
CRP (mg/L)	33.80 ± 5.97	–	–
T28	14.4 ± 1.4	6.9 ± 1.7*	4.8 ± 1.0*
SW28	7.9 ± 0.6	1.6 ± 0.6*	1.1 ± 0.3*
HAQ	1.01 ± 0.15	0.25 ± 0.07*	0.25 ± 0.06*
PGA	7.1 ± 0.6	2.2 ± 0.5*	1.8 ± 0.6*
PHGA	6.7 ± 0.6	2.6 ± 0.5*	1.9 ± 0.5*
DAS28-ESR	6.57 ± 0.23	4.05 ± 0.31*	3.41 ± 0.36*

*$p < 0.05$ compared with week 0.



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Fig. 1. The expression of galectin-9 (Gal-9) in patients with RA. The ACR20/50/70 response rates of patients with RA under Adalimumab treatment, n = 10 per group (A). The volcano map of Gal-9 family expression in PBMCs of Adalimumab treatment cohort (B). The TNF-α related pathways in patients with RA using GO analysis (C). The venn diagram of genes in galectin family and in TNF-α pathways (D). The Gal-9 level in plasma of Adalimumab treatment cohort were tested by ELISA analysis at baseline and week 24, n = 7 per group (E). The expression of Gal-9 mRNA in PBMC of Adalimumab treated patients with RA at baseline and week 24, n = 4 per group (F). The TNF-α (G) and Gal-9 (H) expression was significantly higher in synovial tissues of patients with RA (n = 3) than in osteoarthritis (OA) controls (n = 3) using IHC. Double-staining immunofluorescence of synovial tissues showed the Gal-9 expression (Red) was higher in RA fibroblast-like synoviocytes (vimentin, Green) compared with OA fibroblast-like synoviocytes, n = 3 per group (I). Quantitative results are shown as mean ± SEM. Statistical comparisons between two groups were made using Student's *t*-test. $*p < 0.05$; $**p < 0.01$. ns, not significant.

CO2. The medium was changed every 2 days, and the third to eighth generation of cells were used for the experiments. Extracted RA-FLSs were taken for comparison with OA-FLSs, and purchased RA-FLSs were used for other functional experiments. The cells were then stimulated with different concentrations of TNF-α for 6 h and 24 h separately.

2.3. CIA model and adeno-associated virus (AAV) injection

All animal procedures were performed with the approval of the Institutional Animal Care and Use Committee of Shandong University (Approval no. 23025). Eight-week-old male DBA/1 mice were purchased from Vital River Laboratory Animal Technology (Beijing, China), and housed according to institutional guidelines (temperature: 18–22℃, relative humidity: 50–60 %, noise: 60 dB or less, lighting: 10–14 h, ammonia gas: 20 ppm or less, air change: 8–20 times/hour, air flow: 10–25 cm/min). Bedding was generally changes once/week. Three Gal-9 small interfering (si) RNA molecules were used to knock down Gal-9 expression in mouse NCTC clone 929 fibroblasts (Zhejiang Meissen Cell Technology Co., Ltd.); their sequences are shown as follows: Mus-Gal-9, siRNA-1, GAGGGUAUGUGGUUUGCAATT; siRNA-2, GUCCGAAACACUCAGAUCATT; siRNA-3, GGUCAACACAUGUGUGAAUTT. After the WB assay, the most efficient vector for AAV8-siRNA-3 was constructed (data not shown), and the selected vector structure was pAV-U6-shRNA-CMV-Intron-GFP.AAV8-siGal-9 was delivered into the hind paw by intra-articular injection two weeks before CIA induction. The CIA model was constructed as previously described [25–27]. First, bovine type II collagen (Chondrex, Seattle, WA, USA), and an equal volume of complete Freund adjuvant (Millipore Sigma, Burlington, MA, USA) was mixed well on ice to prepare an emulsion. Eight-week-old male DBA/1 mice were divided into 3 groups (n = 4 mice per group), as follows: control group, CIA group, CIA + AAV8-siGal-9 group. CIA + AAV8-siGal-9 group were injected with AAV8-siGal-9 in the hind paw before CIA was established. After 14 d, on day 0 of model induction, the CIA group and CIA + AAV8-siGal-9 group were injected at the base of the tail with 100 ul of the prepared emulsion. At the same time, the control group was injected an equal volume of PBS using the same methods. To initiate a secondary immune response, a second injection of vector was administered on day 21. At this time clinical arthritis scores were evaluated using a graded scale, as previously described [28]. On day 42, the mice were anesthetized, and tissues were collected for analysis.

2.4. Analysis of RNA-sequencing data

Peripheral blood mononuclear cells (PBMCs) were collected from the above patient cohort and preserved in Trizol. Total RNA from PBMCs was extracted using an RNA Prep Pure Plant Plus Kit (DP441, TIANGEN). All samples have good RNA integrity with RIN numbers between 8 and 10, and libraries were pooled and sequenced using the Illumina Nova Seq machine with 150-bp paired-end sequencing reads. Differentially Expressed Genes (DEGs) were determined using edge R, and genes with $p \leq 0.05$ and $|\log2 \text{ fold change}| \geq 1$ were identified as DEGs.

The purity of the RA-FLSs was determined using NanoPhotometer ® (IMPLEN, CA, USA). Cluster generation and sequencing were performed on a Nova Seq 6000 S4 platform using the Nova Seq 6000 S4 Reagent Kit V1.5. DEG seq was used for differential gene expression analysis between two samples with non-biological replicates. The *p*-value was assigned to each gene and adjusted using Benjamini and Hochberg's approach to control the false discovery rate. Genes with an adjusted *p*-value ≤ 0.05 and fold change ≥ 1.5 or ≤ 0.5 were identified as DEGs.

The expression levels of DEGs were presented as heat and volcano maps using the R packages 'ggplot2′, 'dplyr', and 'pheatmap'. Kyoto Genome Encyclopedia pathway and Gene Ontology (GO) analyses were performed on the DGEs using the R packages 'clusterProfiler', 'topGO', 'Rgraphviz', and 'ggplot2′.

2.5. Enzyme-linked immunosorbent assay

Blood samples were centrifuged at 2000 rpm for 10 min and serum was extracted to test Gal-9 concentration using a Gal-9 Quantikine ELISA kit (Cat no. DGAL90 R&D Systems, USA) according to the manufacturer's instructions. Cell supernatants were collected, centrifuged at 2000 rpm for 5 min and assayed for interleukin-1β (IL-1β), interleukin-6 (IL-6), interleukin-8 (IL-8), chemokine ligand2 (CCL2), matrix metalloproteinases1 (MMP1), matrix metalloproteinases2 (MMP2), matrix metalloproteinases3 (MMP3), matrix metalloproteinases13 (MMP13) using ELISA kits (Multiscience (Lianke) Biotechnology Co., Ltd., Hangzhou, China) according to the manufacturer's instructions.

2.6. Haematoxylin and eosin (HE)

Paw samples were decalcified by immersion in 10 % EDTA until demineralisation, followed by paraffin embedding, tissue wax blocking, and sectioning. Sections (5 mm) were stained with HE.

2.7. Immunohistochemistry (IHC) and immunofluorescence (IF)

IHC staining was performed according to a standard staining procedure [29].The following primary antibodies were used: anti-Galectin-9 antibody (1:200; Abcam, UK), anti-MAFB antibody (1:100; Affinity, China), and anti-TNF-α antibody (1:200; Immunoway, China).

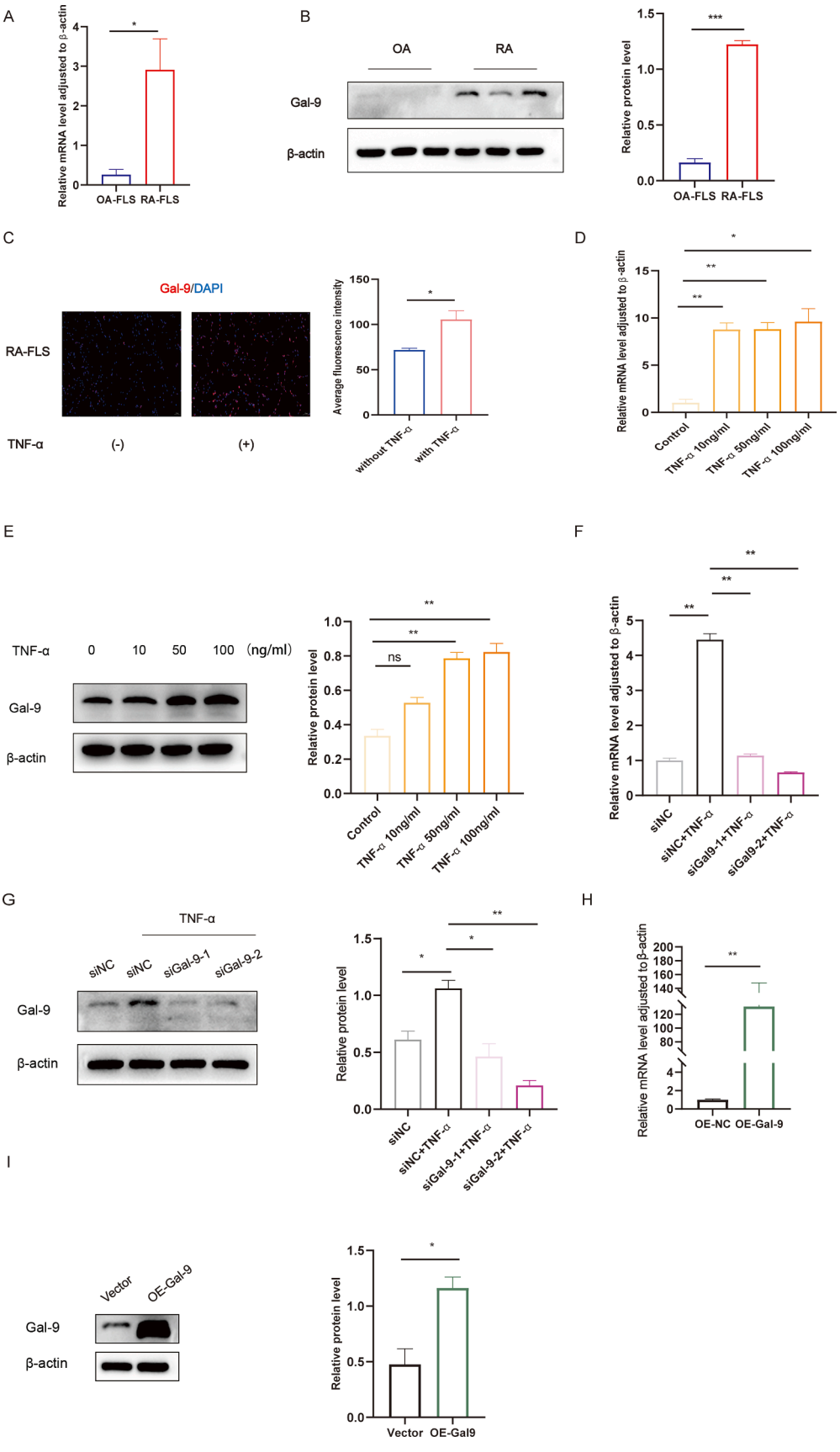
IF staining was performed according to a standard staining procedure [30].The following antibodies were used: anti-Galectin-9 antibody (1:200; Abcam, UK) and anti-TNF-α antibody (1:200; Immunoway, China). Alexa Fluor ™ 488 anti-vimentin antibody (677809; Bio-Legend), and Donkey Anti-Sheep IgG NorthernLights™ NL557-conjugated Antibody (NL010; R&D Systems) were also used.

2.8. Transfection of siRNA and plasmid

For gene silencing, appropriate cells were seeded in 6-well plates, siRNA (Gene Pharma, Shanghai, China) was transfected into RA-FLSs using the Jet PRIME transfection reagent (Polyplus, France). For gene overexpression studies, RA-FLSs were transfected with the eukaryotic expression vector containing the Gal-9 gene, using Jet PRIME transfection reagent (Polyplus, France). An empty vector was used as the control. Sequence of siRNA are as follows: Homo-Gal-9, siRNA-1,GUGCAGAGCUCAGAUUUCATT; siRNA-2, GUCAGCACCUGUUUGAAUATT; Homo-MAFB,siRNA-1, CCAUCACCAUCAUCACCAA.

2.9. RNA extraction and real-time qPCR

Total RNA was isolated from RA-FLSs using RNA fast 200 (Fastagen, Shanghai, China). RNA was reverse-transcribed into cDNA using a



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Fig. 2. Gal-9 expression was elevated in RA synovial tissues and TNF-α-induced- fibroblasts (FLSs). Expression levels of Gal-9 of RA-FLSs (n=3 per group) and OA-FLSs (n=3 per group) detected using qPCR (A) and western blot (B). Representative images of immunofluorescence of RA-FLSs stimulated by TNF-α (50ng/ml) for 24 h. Scale bar, 100μm. (C). Expression levels of Gal-9 were illustrated with different concentrations of TNF-α stimulation (0, 10, 50, 100 ng/ml) in RA -FLSs detected using qPCR (D) and western blot (E). Gal-9 mRNA and protein expression levels in TNF-α (50ng/ml)-induced RA-FLSs after Gal-9 knockdown using qPCR (F) and western blot(G). Relative mRNA and protein levels of Gal-9 after Gal-9 overexpression (OE- Gal-9) in TNF-α (50ng/ml)-induced RA-FLSs using qPCR (H)and western blot (I). Quantitative results are shown as mean ± SEM, and the cell experiments were repeated at least three times. Statistical comparisons between two groups were made using Student's *t*-test. Multiple group comparisons were performed using one-way analysis of variance (ANOVA) followed by Dunnett's for multiple comparisons. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. ns, not significant. Values of $p < 0.05$ were considered significant.

reverse transcription kit (AG, Accurate Biotechnology). The cDNA was amplified using a SYBR Green PCR kit (AG, Accurate Biotechnology) according to the manufacturer's instructions. The primer sequences for the target genes are listed as follows: Homo-β-actin Forward GAAGAGCTACGAGCTGCCTGA; Reverse CAGACAGCACTGTGTTGGCG; Homo-Gal-9 Forward CCCGAGGAGAGGAAGACACAC; Reverse CCCGTTCACCATCACCTTGAAATC; Homo-PHLDA2 Forward GACAGCCTCTTCCAGCTATGG; Reverse TCGATCTCCTTGTGGTCGGT; Homo-CYP1B1 Forward AACGTACCGGCCACTATCAC; Reverse TCACCCATACAAGGCAGACG; Homo-STING1 Forward TACAACAACCTGCTACGGGG; Reverse TCTGCTGGGGCAGTTTATCC; Homo-MAFB Forward CTCAGCACTCCGTGTAGCTC; Reverse CGGGGTTCATCTGCTGGTAG; Homo-NOLC1 Forward GGGCTCTCCAAGACCACAAG; Reverse ACTACCACTGCCGCCTTTTT; Homo-TPM4 Forward TGACCGCAAATACGAGGAGG; Reverse TAGTTCAGACACCTCCGCAC; Homo-ZNF267 Forward TGTCTCTAAGCCGGACCTGA; Reverse AGCTCCCATGTCTCCTCGAT; Homo-MLCK Forward CATCTCTGCGGGTTACGAGG; Reverse TTCTTCACGTCCTCCCGGT; Mus-β-actin Forward GGCTGTATTCCCCTCCATCG; Reverse CCAGTTGGTAACAATGCCATGT; Mus-Gal-9 Forward ATGGGCTTTACCCGTCCAAG; Reverse GCATCCTCCCAAGCAGACTT. Gene expression was calculated by the ΔΔCt method. β-actin was used as an internal control.

2.10. Western blot analysis (WB)

WB detection of RA-FLSs were perform according to a standard staining procedure [52]. After cell lysis and centrifugation, the protein concentrations in the supernatants were measured with the BCA Protein Assay Kit. After preparation of the required protein concentration, the samples were electrophoresed in an appropriate separation gel under 150-V constant voltage and transferred onto a polyvinylidene fluoride membrane (Immobilon-P; Millipore). The membrane was blocked with 5 % skim milk in a shaker for 1 h at room temperature, and incubated with a primary antibody overnight at 4 °C. The membrane was then washed with Tris-buffered saline containing 0.1 % Tween-20 and incubated with an appropriate secondary antibody for 1 h at room temperature. The blots were established via an enhanced chemiluminescence (ECL kit, Millipore). The following primary antibodies were used: anti-β actin(1:1000, ZSGB-BIO,TA-09, China), anti-phosphorylated (p)-PI3K(1:1000,CST,#4228,USA), anti-PI3K(1:1000, CST,#4249,USA), anti-p-AKT(1:1000,CST,#4060,USA), anti-AKT (1:1000,CST,#9272,USA), anti-p-mTOR(1:1000,CST,#5536,USA), anti-mTOR(1:1000,CST,#2972,USA), anti-MAFB(1:1000,CST,#79737, USA), anti-Bax(1:1000,CST,#2772,USA), anti-Bcl-2(1:1000,CST, #15071,USA), anti-Caspase3(1:1000,CST,#9662,USA), anti-Caspase7 (1:1000,CST,#12827,USA), anti-Caspase9(1:1000,CST,#9504,USA), anti-NOLC1 (1:1000,ABclonal,A21134, China); anti-Galectin-9(1:1000, Abcam, ab69630,UK).

2.11. Cell proliferation assay

Cell proliferation was detected using a Cell Counting Kit-8 (Dojindo Molecular Technologies) in accordance with the manufacturer's instructions. Cells (7×10^3) were seeded into the wells of a 96-well plate and evaluated at 0 h, 24 h, 48 h, 72 h after cell adherence. Following addition of 10 μl of CCK-8 solution to each well, the absorbance was measured at 450 nm with a microplate reader (ELX-808, BioTek, USA). The EdU incorporation assay was performed with a Cell-Light EdU DNA Cell Proliferation Kit (RiboBio, Guangzhou, China) following the manufacturer's instructions. The experiments were repeated three times.

2.12. Cell apoptosis

Pretreated RA-FLSs ($3–4 \times 10^5$) in 6-well plates were digested with 0.25 % trypsin (BC-CE-005; Bio-Channel) and centrifuged to collect the cell pellets. Apoptosis was detected via flow cytometry using an FITC Apoptosis assay kit (Vazyme, China) according to the manufacturer's instructions. The excitation wavelength of the flow cytometer was 488 nm, the green fluorescence of FITC was detected in the FL1 channel, and the red fluorescence of PI was detected in the FL3 channel, and 10,000 events were collected for each sample. The data were analyzed using Flow Jo 18.0 software, with FL1 as the horizontal coordinate, and FL3 as the vertical coordinate, and the negative/positive boundaries of the two fluorescence parameters were determined based on the values of the fluorescence of FITC and PI, and the cross-gate was delineated. Cls in the experiment can be divided into three subpopulations, live cells are double negative (Annexin V-FITC-/PI-); early apoptotic cells are (Annexin V-FITC+/PI-); late apoptotic cells are (Annexin V-FITC+/PI +).

The pretreated RA-FLSs in 6-well plates were fixed in 4 % paraformaldehyde. The TUNEL assay (Beyotime Biotechnology) was performed according to the manufacturer's instructions. Finally, the slides were visualised under a fluorescence microscope (Nikon Eclipse C1, Japan).

2.13. Migration and invasion assay

For cell migration analysis, $6–8 \times 10^4$ cells suspended in serum-free medium were inoculated into the upper chamber of 8 μm wells and cultured in DMEM with 20 % fetal bovine serum in the lower chamber of 24-well plates (Corning #3378, USA). After 48 h of routine culture, the chamber was removed and a cotton swab was used to gently wipe the cells in the upper chamber. Next, 0.1 % crystal violet solution (G1063; Solarbio) was added to stain the cells for 10 min. After air drying, the cells were observed and counted in five fields under a forward fluorescence microscope (Leica DM2500).

For the invasion assay, the upper chamber was coated with 200 μg/mL Matrigel (Solarbio, China) and 6×10^4 cells were seeded into the upper chamber. Other procedures were similar to those used for the migration assays. All experiments were repeated at least three times.

2.14. Co-immunoprecipitation assay

The pretreated cells were washed with PBS and lysed with cell lysis buffer containing protease inhibitors for 30 min on ice. After centrifugation at 12,000g for 15 min at 4 °C, the supernatant was collected and incubated with Gal-9 antibody (2ug, ab69630, Abcam) and MAFB antibody (1:50, #79737, Cell Signaling Technology) or rabbit normal IgG control (Beyotime Biotechnology, China) by spinning overnight at 4 °C, followed by further incubation with 30 μl of pre-washed protein A/G magnetic beads for 2 h at 4 °C. After incubation, the immune complexes were washed three times with lysis buffer, and the proteins were eluted by adding SDS sample buffer and incubated at 100 °C for 5 min. The eluates were analysed via WB using the primary antibodies described

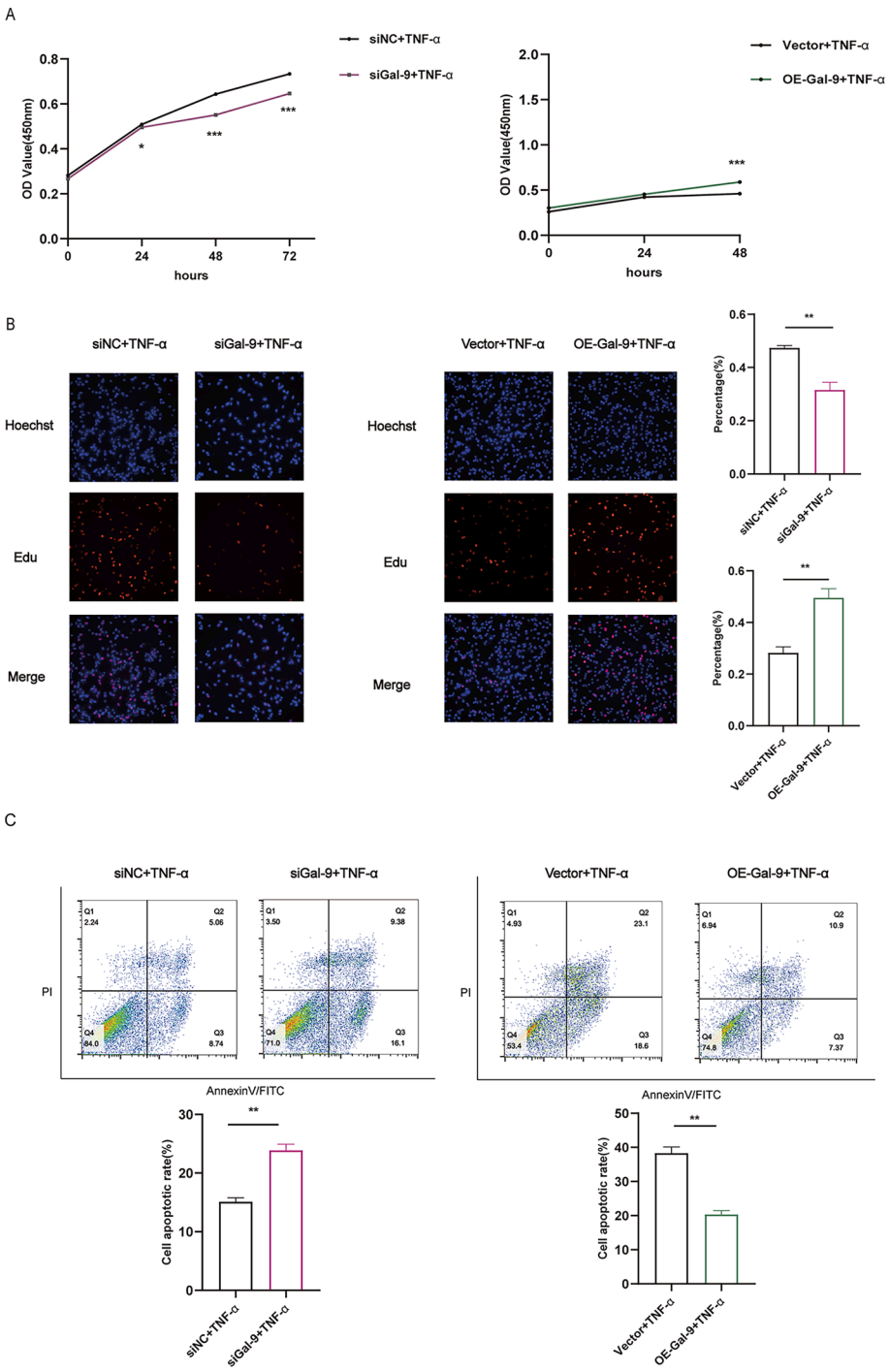


Fig. 3. Both Gal-9 knockdown and overexpression influenced the TNF-α induced RA-FLSs activation *in vitro*. Cell growth of RA-FLSs was analyzed via CCK8 assay with TNF-α (50 ng/ml) stimulation in Gal-9 knockdown and negative control group (A). RA-FLSs cell proliferation was observed through Edu assay (B). RA-FLSs cell apoptosis was analyzed using FACS staining with Annexin V-FITC and PI (C) and Tunel staining (D). Biomarkers of cell apoptosis were tested using western blot (E). Migration and invasion were detected using transwell assays (F). ELISA analysis of IL-1β, IL-6, IL-8, CCL2, MMP1, MMP2, MMP3, and MMP 13 levels in supernatant of cultured FLSs with Gal-9 knockdown / overexpression (G). Quantitative results are shown as mean ± SEM, and the cell experiments were repeated three times. Statistical comparisons between two groups were made using Student's *t*-test. Multiple group comparisons were performed using one-way analysis of variance (ANOVA) followed by Dunnett's for multiple comparisons. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. ns, not significant. Values of $p < 0.05$ were considered significant.

above.

2.15. Statistical analysis

All experiments were conducted independently at least three times. Data were analyzed and plotted using the GraphPad Prism 8 software. Data were expressed as mean ± SEM. Statistical comparisons between two groups were made using Student's *t*-test. Multiple group comparisons were performed using one-way analysis of variance (ANOVA) followed by Dunnett's for multiple comparisons.

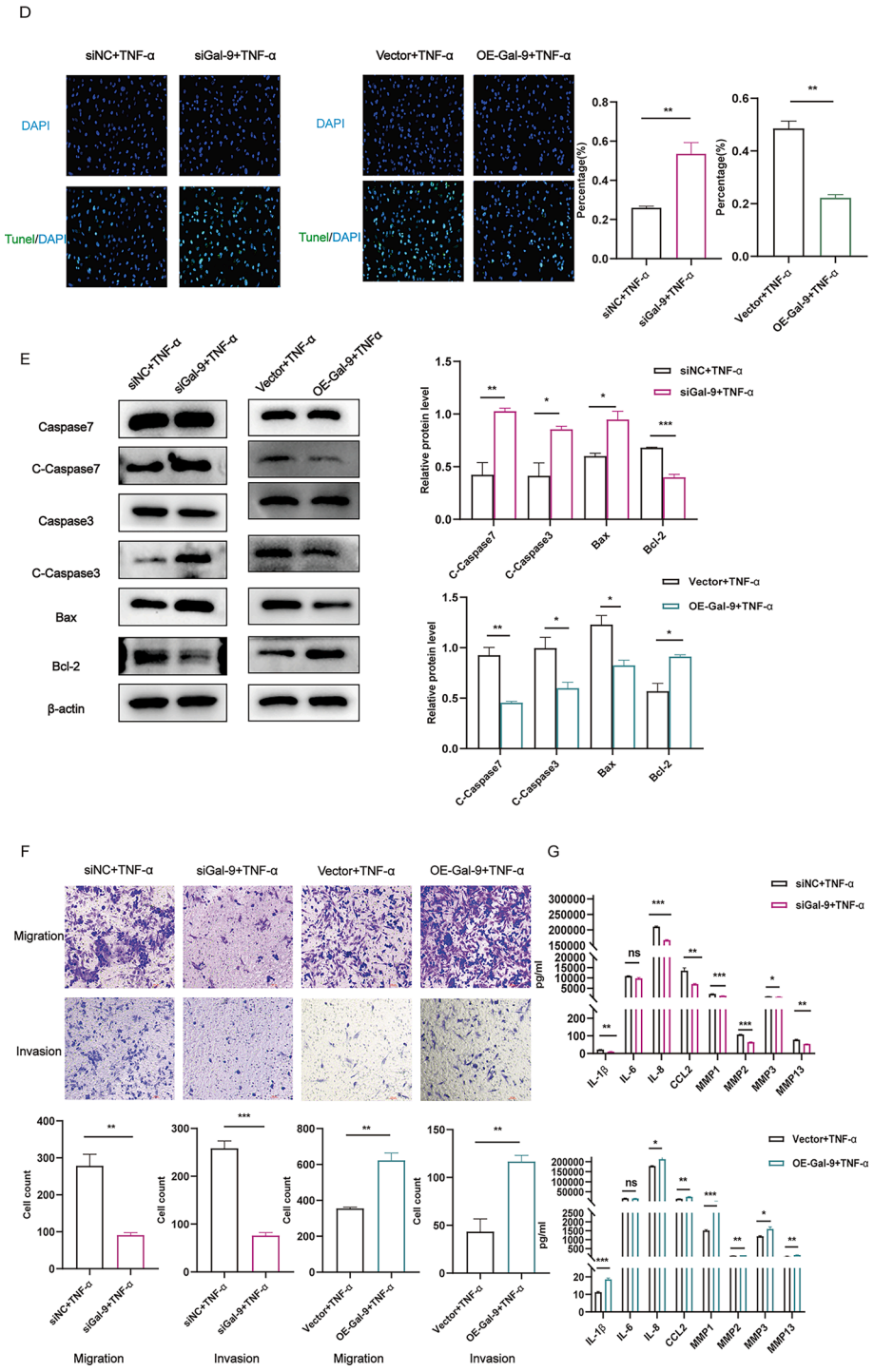
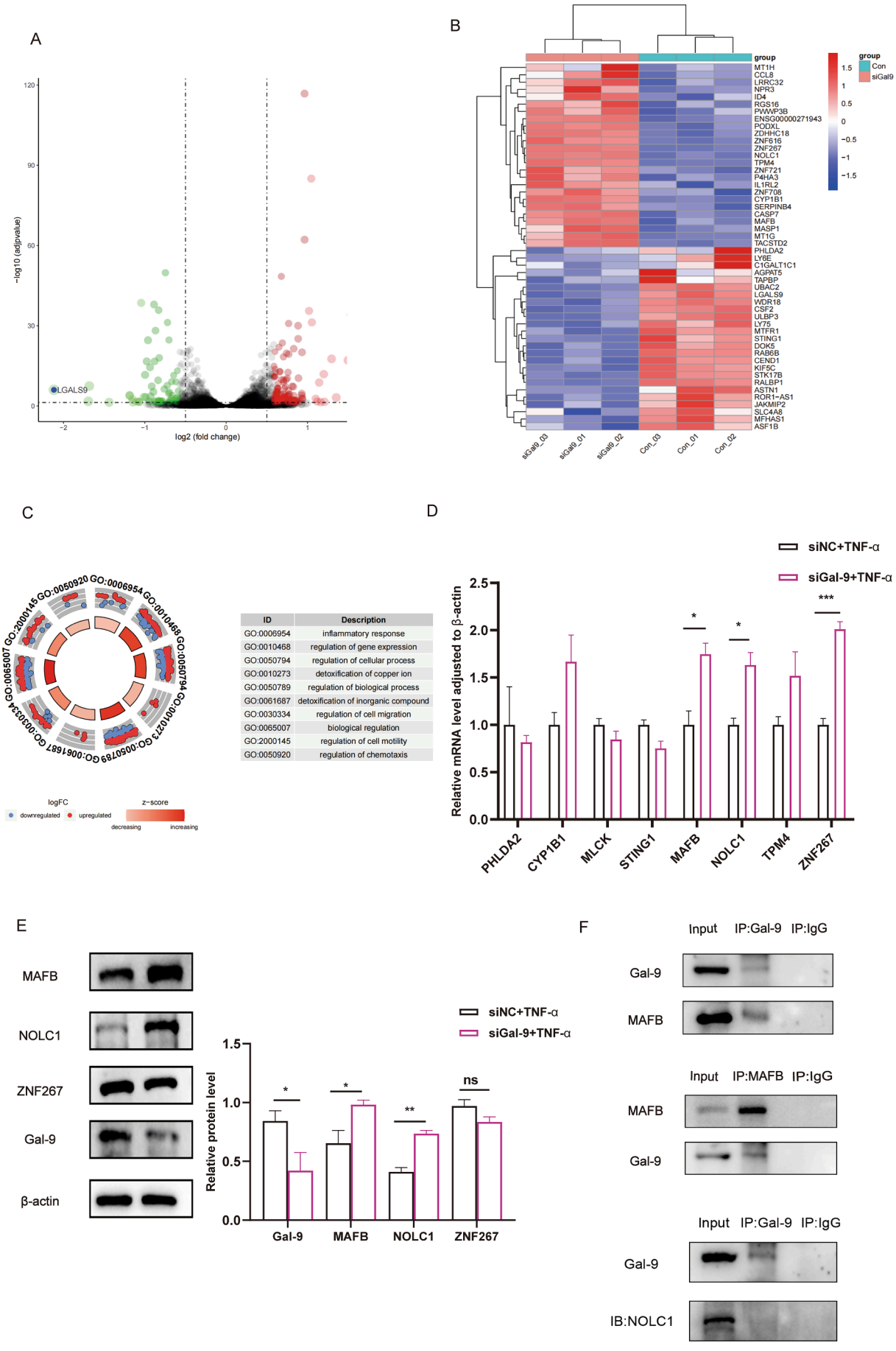


Fig. 3. (*continued*).

3. Results

3.1. Expression of the galectin family in the Adalimumab treatment cohort

We included 10 patients with RA of high disease activity. The patients treated with Adalimumab and methotrexate (MTX) for 6 months showed considerable improvements in the joint assessment, health assessment questionnaire (HAQ), and disease activity at 3 and 6 months compared to baseline (Table 1). The same trend was observed for the American College of Rheumatology (ACR) 20, ACR 50, and ACR 70 treatments (Fig. 1A). The mRNA levels of peripheral blood mononuclear cells (PBMCs) in patients substantially changed after adalimumab treatment, with 1394 differentially expressed genes (DEGs). The expression of Gal-1, Gal-2, and Gal-9 of the galectin family considerably decreased after Adalimumab treatment (Fig. 1B). We further focused on the most direct-related pathways that were significantly altered after Adalimumab treatment (Fig. 1C). Gal-9 is the only galectin involved in the regulation of important TNF-related pathways (Fig. 1D), and we next investigated exactly how Gal-9 is involved in TNF-induced activation of FLSs. Enzyme-linked immunosorbent assay (ELISA) results showed that Gal-9 levels in the plasma significantly decreased after Adalimumab



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Fig. 4. The screening for target genes with Gal-9 knockdown in RA-FLSs *in vitro*. The volcano map of RA-FLSs with Gal-9 knockdown (A). The heat map of the top 50 DEGs in the RA-FLSs with down-regulated expression of Gal-9 (B). The top 10 pathways in the RA-FLSs, with GO analysis for the DEGs (C). The mRNA expression of eight target genes in TNF-α stimulation RA-FLSs with depletion of Gal-9 via qPCR (D). MAFB, NOLC1, ZNF267 protein expression in TNF-α activated RA-FLSs with depletion of Gal-9 using WB (E). CO-IP was performed to detect Gal-9-MAFB and Gal-9-NOLC1 co-expression in RA-FLSs (F). Quantitative results are shown as mean ± SEM, and the cell experiments were repeated at least 3 times. Statistical comparisons between two groups were made using Student's *t*-test. Multiple group comparisons were performed using one-way analysis of variance (ANOVA) followed by Dunnett's for multiple comparisons. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. ns, not significant. Values of $p < 0.05$ were considered significant.

treatment for 6 months compared to baseline levels in seven patients with RA(Seven patients contributed their serum and 3 others refused), ($p < 0.01$) (Fig. 1E). The mRNA levels of Gal-9 in PBMCs also showed a decreasing trend after Adalimumab treatment in the four patients with RA (only 4 RNA samples remained after RNA sequencing). ($p = 0.08$) (Fig. 1F).

3.2. TNF-α stimulates the expression of Gal-9 in RA-FLSs

TNF-α and Gal-9 expression was increased in RA synovium compared to OA synovium according to immunohistochemical (IHC) results ($p < 0.01$) (Fig. 1G, H) and Gal-9 expression was shown to be elevated in RA-FLSs compared to OA-FLSs via immunofluorescence (IF) ($p < 0.01$) (Fig. 1I). Gal-9 expression was upregulated in RA-FLSs, as confirmed by qPCR ($p < 0.05$) (Fig. 2A) and western blotting (Fig. 2B) ($p < 0.001$). Interestingly, stimulation of RA-FLSs with TNF-α resulted in stronger Gal-9 fluorescence as detected by IF ($p < 0.05$) (Fig. 2C). Subsequently, stimulating RA-FLSs with different concentrations of TNF-α (10, 50, and 100 ng/ml) for 6 h, the results suggested that the stimulation of TNF-α could promote the elevation of Gal-9 mRNA expression ($p < 0.01$) (Fig. 2D), and there was no statistical difference between the three different concentration groups when comparing with each other. Stimulating RA-FLSs with different concentrations of TNF-α (10, 50, and 100 ng/ml) for 24 h, the results showed that the stimulation of TNF-α could enhanced the expression of Gal-9 protein ($p < 0.01$; Control group vs. TNF-α 10 ng/ml group, $p = 0.0607$;Control group vs. TNF-α 50 ng/ml group, $p = 0.0032$;Control group vs. TNF-α 100 ng/ml group, $p = 0.0062$;TNF-α 10 ng/ml group vs. TNF-α 50 ng/ml group, $p = 0.0171$; TNF-α 10 ng/ml group vs. TNF-α 100 ng/ml group, $p = 0.0340$;TNF-α 50 ng/ml group vs. TNF-α 100 ng/ml group, $p = 0.9216$). Similarly, we also stimulated FLSs with different concentrations of TNF-α for 6 and 12 h, unfortunately, after TNF-α stimulation, the expression of Gal-9 protein was not significantly different between different concentration groups (data not shown). Therefore, in the following experiments, 50 ng/ml of TNF-α stimulated for 24 h was used. The results suggested that the appropriate concentration of TNF-α stimulated for an appropriate period of time led to an increase in Gal-9 production in RA-FLSs.

3.3. Knockdown of Gal-9 inhibited the activation of RA-FLSs with TNF-α stimulation

We investigated the biological function of Gal-9 in TNF-α induced RA-FLSs. Small interfering RNA (siRNA) (Fig. 2F, G) and the pcDNA3.1 overexpression vector (Fig. 2H, I) for Gal-9 were transfected into TNF-α induced RA-FLSs and the efficiencies were verified. Cell Counting Kit-8 (CCK-8) assays demonstrated that Gal-9 siRNA inhibited the proliferation of TNF-α induced RA-FLSs (Fig. 3A), Edu assays confirmed the same findings (average 123/261,47.4 %) in siNC group and (average 42/134,31.6 %) in siGal-9 group, $p < 0.01$.(Fig. 3B). Flow cytometry results showed that the proportion of apoptotic cells was elevated in the downregulation of Gal-9 group compared to that of the small interfering RNA of negative control (siNC) group (average15.12 % in siNC group and average 23.88 % in siGal-9 group, $p < 0.01$) in TNF-α induced RA-FLSs (Fig. 3C). Similarly, the proportion of positive cells for Tunel staining was increased in the siGal-9 group, (average 39/152, 25.9 %) in siNC group and (average 82/159, 53.6 %) in siGal-9 group, $p < 0.01$ (Fig. 3D). During Gal-9 interference, the cleavage of apoptotic protein, caspase-3/7 was enhanced, pro-apoptotic protein Bax was increased, and anti-apoptotic protein Bcl-2 was decreased (Fig. 3E). Futher, transwell assays were performed to assess whether the downregulation of Gal-9 affected cell migration and invasion. Knockdown of Gal-9 impaired the migration and invasion of TNF-α induced RA-FLSs (Fig. 3F). Several studies in recent years found that TNF-α stimulated RA-FLSs to produce a variety of inflammatory factors, such as IL-1-β, IL-6, IL-8, CCL2 and chemokines MMP1, MMP2, MMP3, MMP13 [30–33], which are involved in TNF-α induced activation of FLSs. As shown in Fig. 3G, silencing Gal-9 reduced the secretion of interleukin (IL)-1β, IL-8, chemokine ligand 2 (CCL2), matrix metalloproteinases 1 (MMP1), metalloproteinases 2 (MMP2), metalloproteinases 3 (MMP3) and matrix metalloproteinases 13 (MMP13), nevertheless, the reduction of Gal-9 did not decrease the expression level of IL-6 in cell supernatants.

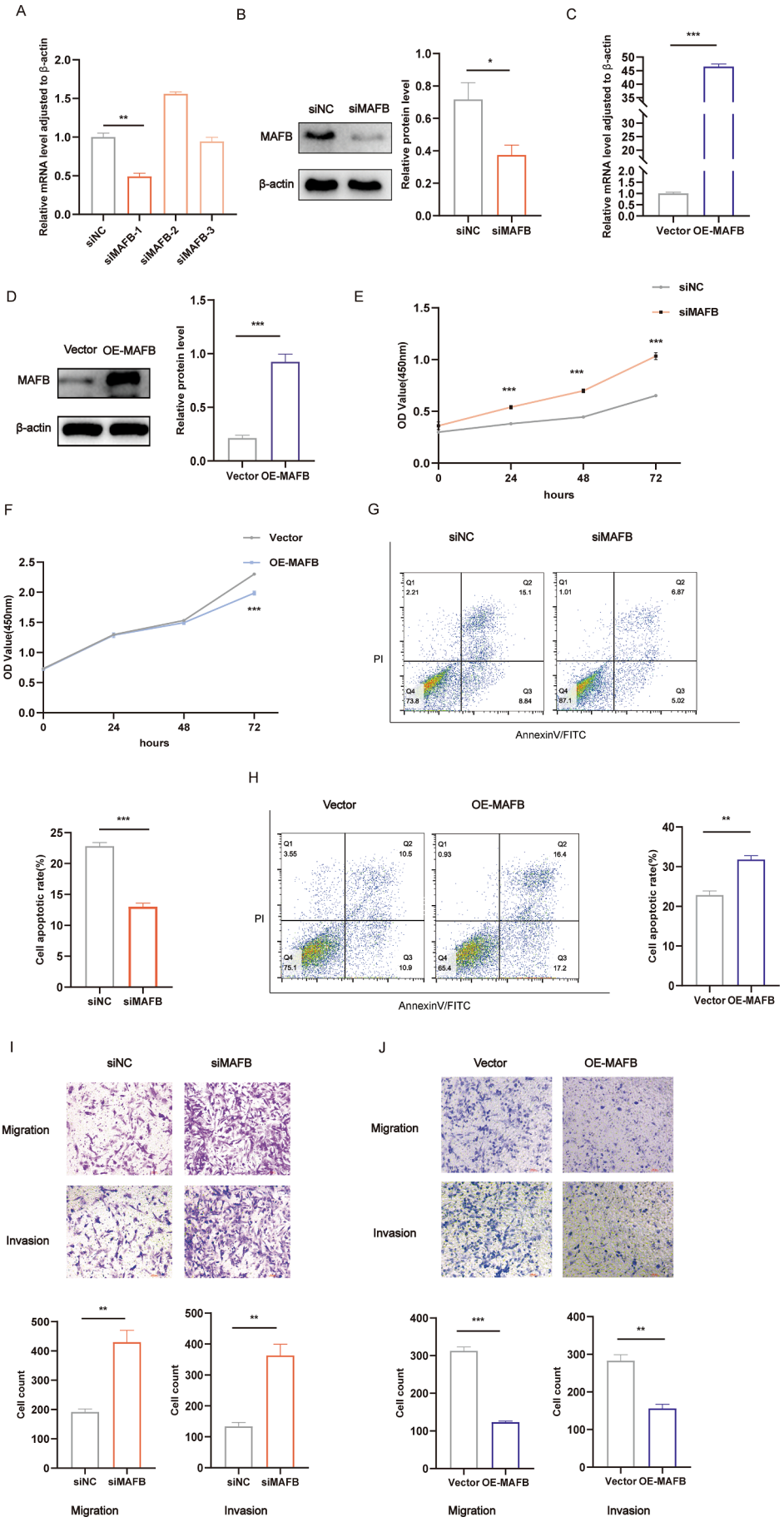
Conversely, the overexpression of Gal-9 promoted the proliferation, migration and invasion of TNF-α-induced RA-FLSs (Fig. 3A, B, F), and increased the production of IL-1β, IL-8, CCL2, MMP1, MMP2, MMP3, and MMP13 (Fig. 3G), whereas flow cytometry revealed that overexpression of Gal-9 significantly decreased the proportion of apoptotic cells compared with the control group ($p < 0.01$) (Fig. 3C). The proportion of Tunel staining positive cells was decreased compared with the control group ($p < 0.01$) (Fig. 3D). The results suggested that knockdown of Gal-9 suppressed activation and facilitated apoptosis of FLSs, reducing the production of inflammatory cytokines and chemokines, whereas overexpression of Gal-9 revealed the opposite phenomenon.

3.4. Identification of MAFB as a target for Gal-9 in RA-FLSs activation

A total of 180 DEGs were found in Gal-9-knockdown RA-FLSs compared to the control, with 57 downregulated and 123 upregulated genes (Fig. 4A). The heat map shows the top 50 genes with a fold-change in the absolute value (Fig. 4B). Gene ontology (GO) analysis suggested that these genes were mainly involved in inflammatory response, regulation of cellular processes, regulation of biological processes, regulation of cell migration, biological regulation, regulation of cell motility, and regulation of chemotaxis (Fig. 4C). We selected 8 genes (*PHLDA2*, *CYP1B1*, *MLCK*, *STING1*, *MAFB*, *NOLC1*, *TPM4*, and *ZNF267*) that may be involved in Gal-9 deregulation for further validation. MAFB, NOLC1 and ZNF267 mRNA expression levels were significantly elevated in the Gal-9 knockdown group (Fig. 4D). Genes with significant differences were validated through western blot (WB), Gal-9 knockdown resulted in a more substantial elevation of MAFB ($p < 0.05$) and NOLC1 ($p < 0.01$) in protein levels than ZNF267 ($p = 0.12$) (Fig. 4E). To further characterise the relationship between Gal-9 and MAFB and NOLC1, co-immunoprecipitation (Co-IP) assays were performed to confirm the relationship between Gal-9, MAFB, and NOLC1. Fig. 4F showed that endogenous Gal-9 forms a complex with MAFB. However, no binding of Gal-9 protein and NOLC1protein was identified via Co-IP (Fig. 4F).

3.5. Overexpression of MAFB inhibited the activation of RA-FLSs with TNF-α stimulation

The data obtained using qPCR and WB showed the efficiency of siRNA silencing MAFB gene (Fig. 5A, B) and plasmid transfection overexpressing MAFB genes (Fig. 5C, D). Notably, MAFB silencing promoted cell proliferation while MAFB overexpressing diminished cell proliferation, as indicated using the CCK-8 assays in RA-FLSs (Fig. 5E, F). Conversely, the rates of apoptotic cells was reduced in the RA-FLSs transfected with siMAFB compared to that of the siNC group



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Fig. 5. MAFB was necessary for the inhibitory role of Gal-9 in RA-FLSs *in vitro*. Relative mRNA and protein levels of MAFB after MAFB knockdown in RA-FLSs via qPCR (A) and western blot (B). Relative mRNA and protein levels of MAFB overexpression through qPCR (C) and western blot (D). Cell growth was analyzed by CCK8 assay in MAFB knockdown/overexpression in RA-FLSs in comparison with siNC and vector groups (E, F). Cell apoptosis was analyzed via staining with Annexin V-FITC and PI (G, H). Migration and invasion were detected by transwell migration and invasion assays (I, J). Quantitative results are shown as means ± SEM, and the cell experiments were repeated at least 3 times. Statistical comparisons between two groups were made using Student's *t*-test. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. Values of $p < 0.05$ were considered significant.

(proportion of apoptotic cells, 13.03 % in the siMAFB group and 22.82 % in the siNC group, $p < 0.001$) (Fig. 5G), the opposite tend was showed in MAFB overexpression group (22.86 % in the Vector group and 31.8 % in the MAFB overexpression group, $p < 0.01$) (Fig. 5H). The number of cell migration and invasion in the siMAFB group was greater than that in the siNC group as detected using the Transwell assay (Fig. 5I), whereas overexpression of MAFB could weaken the migration and invasion of RA-FLSs (Fig. 5J).

3.6. siGal-9 inhibits apoptosis and secretion of inflammatory factors via the PI3K-AKT signaling pathway in TNF-α induced RA-FLSs

Our results revealed that Gal-9 interference decreased the ratio of PI3K, AKT, and mTOR phosphorylated proteins to total proteins in the TNF-α-induced RA-FLSs, whereas the opposite result was obtained with Gal-9 overexpression (Fig. 6A). In addition, we examined the phosphorylation status of key proteins of other common pathways (STAT3, p65, EKR, p38, and JNK) using the WB method (data not shown); however, no considerable changes in phosphorylated proteins were observed. Rescue experiments were performed using small molecule activators of the PI3K/AKT/mTOR signaling pathway, SC79(AKT agonist) and 740YP (PI3K agonist). The apoptotic effect via Gal-9 knockdown in TNF-α induced RA-FLSs was significantly weakened (siGal-9 group 30.14 % vs siGal-9 + SC79 group 24.98 %, $p < 0.01$; siGal-9 group 30.14 % vs siGal-9 + 740YP group 24.79 %, $p < 0.05$), when we added SC79 (AKT agonist) and 740YP (PI3K agonist) to the culture medium (Fig. 6B). The release of cytokines increased after the addition of pathway agonists compared with the knockdown Gal-9 group (Fig. 6C). This result highlights the PI3K-AKT pathway as an important influencing factor in TNF-α induced RA-FLSs when silencing Gal-9.

3.7. Local knockdown of Gal-9 by adeno-associated virus (AAV)-siGal-9 attenuated the development of CIA

To determine whether Gal-9 deficiency alleviates the development of inflammatory arthritis, we established a CIA model in which AAV-siGal-9 was delivered into the hind paw by intra-articular injection two weeks before CIA induction. The other steps followed the CIA modeling schematic (Fig. 7A). In comparison with the CIA model, photographs of hind paw showed that the joints and paws of the AAV-siGal-9 group exhibited less swelling than the CIA group (Fig. 7B), and arthritis scores were lower than those of the CIA group (Fig. 7C). The metatarsophalangeal joints and paws of mice were collected at day 42. Histological analysis was performed and severe synovial hyperplasia and inflammatory cell infiltration in the CIA model were found to be diminished by AAV-siGal-9 treatment (Fig. 7D). IHC analysis showed increased Gal-9 expression and decreased MAFB expression in the CIA group compared to the AAV-siGal-9 group (Fig. 7E, F). In summary, these experiments suggest that AAV-siGal-9 plays a protective role in the CIA model.

4. Discussion

Gal-9 is a β-galactoside binding lectin (34–39 kDa) encoded by the *LGALS9* gene [17], that is expressed in several human organs and tissues [34]. It is involved in cell adhesion and communication [35,36], regulation of the immune response, and cell differentiation and death [37–40], more functions are still being explored. Gal-9 expression in synovial tissue and synovial fluid was considerably higher in patients with RA compared to patients with OA [41]. Gal-9 knockdown promoted apoptosis in RA-FLSs, delaying the progression of RA [18]. Our previous study showed that Gal-9 promoted angiogenesis and monocyte migration *in vitro* and induces acute inflammatory arthritis in mice [42]. These results suggest that Gal-9 plays a positive role in promoting RA progression. In addition, some clinical cohort studies suggesting a correlation between Gal-9 levels and disease activity in patients with RA [20,43], showing the same trend as in our previous study, which also suggested that baseline Gal-9 levels may be predictive of clinical treatment response [15,16,44]. These results indicate that Gal-9 plays an important role in pathogenesis of RA. However, the mechanism by which Gal-9 mediates RA inflammation and immune response requires further investigation.

Our study revealed that Gal-9 were highly expressed in the synovium and RA-FLSs of patients with RA. Gal-9 serum levels were significantly decreased in patients with RA after treatment with adalimumab. Transcriptome sequencing showed that Gal-9 is involved in the TNF-α pathway. Knockdown of Gal-9 inhibits proliferation, migration, and inflammatory response via MAFB and PI3K-AKT pathway in TNF-α activated FLSs. This study is an original proposal supporting the finding that Gal-9 is involved in the TNF-α pathway, thus opening a new avenue for the treatment of RA. Some investigators had verified that exogenous Gal-9 promotes apoptosis of FLSs [41], whereas endogenous Gal-9 inhibited apoptosis [18]; however, these studies did not elucidate the specific mechanism. We found that interfering with Gal-9 facilitates apoptosis through the MAFB and PI3K-AKT pathway in TNF-α induced RA-FLSs. A new publication in 2023 found that Gal-9 expression was considerably higher in the joint fluid compared to the serum in RA patients, that CCL2 and IL-6 secretion by RA-FLSs were reduced after the addition of antibodies that neutralise Gal-9 *in vitro*, suggesting that Gal-9 is mainly produced locally in arthritis and regulates RA-FLSs activity through sustained expression, thereby increasing the level of inflammation in the synovial microenvironment [20]. In our study, downregulation of Gal-9 was able to reduce the secretion of CCL2, which was essential for regulating the interaction between FLSs and inflammatory cells and stabilising the local synovial microenvironment. Gal-9 activated polymorphonuclear leukocytes *in vitro*, stimulates cytokine secretion, and promotes cell migration and survival [45]. We observed that enhancing the endogenous overexpression of Gal-9 could strengthen FLSs migration, proliferation, and IL-1 and IL-8 secretion with TNF-α stimulation. This might be a potentially important mechanism in the development of RA. Some studies had reported that Gal-9 contributes to cell proliferation [46], while others had reported that it induced apoptosis [47]. In atopic dermatitis, exogenous Gal-9 co-stimulated keratin-forming cells with TNF-α/IFN-γ, which inhibited the migration of keratin-forming cells [48]. Another study showed that endogenous overexpression of Gal-9 suppressed the proliferation and migration of hepatocellular carcinoma cells and promoted apoptosis and that the expression level of Gal-9 in hepatocellular carcinoma tissues was lower than that in adjacent normal tissues[49]. RA-FLSs are tumour-like cells, but our findings are inconsistent with those in hepatocellular carcinoma: Gal-9 expression in the synovium and FLSs of patients with RA was considerably higher than that of patients with OA, and the deletion of Gal-9 injured proliferation, migration, and promoted the apoptosis in TNF-α induced RA-FLSs. Therefore, Gal-9 has pro- and anti-inflammatory functions depending on the amount of expression and cellular location. These data confirmed the important role of Gal-9 in the onset and progression of RA.

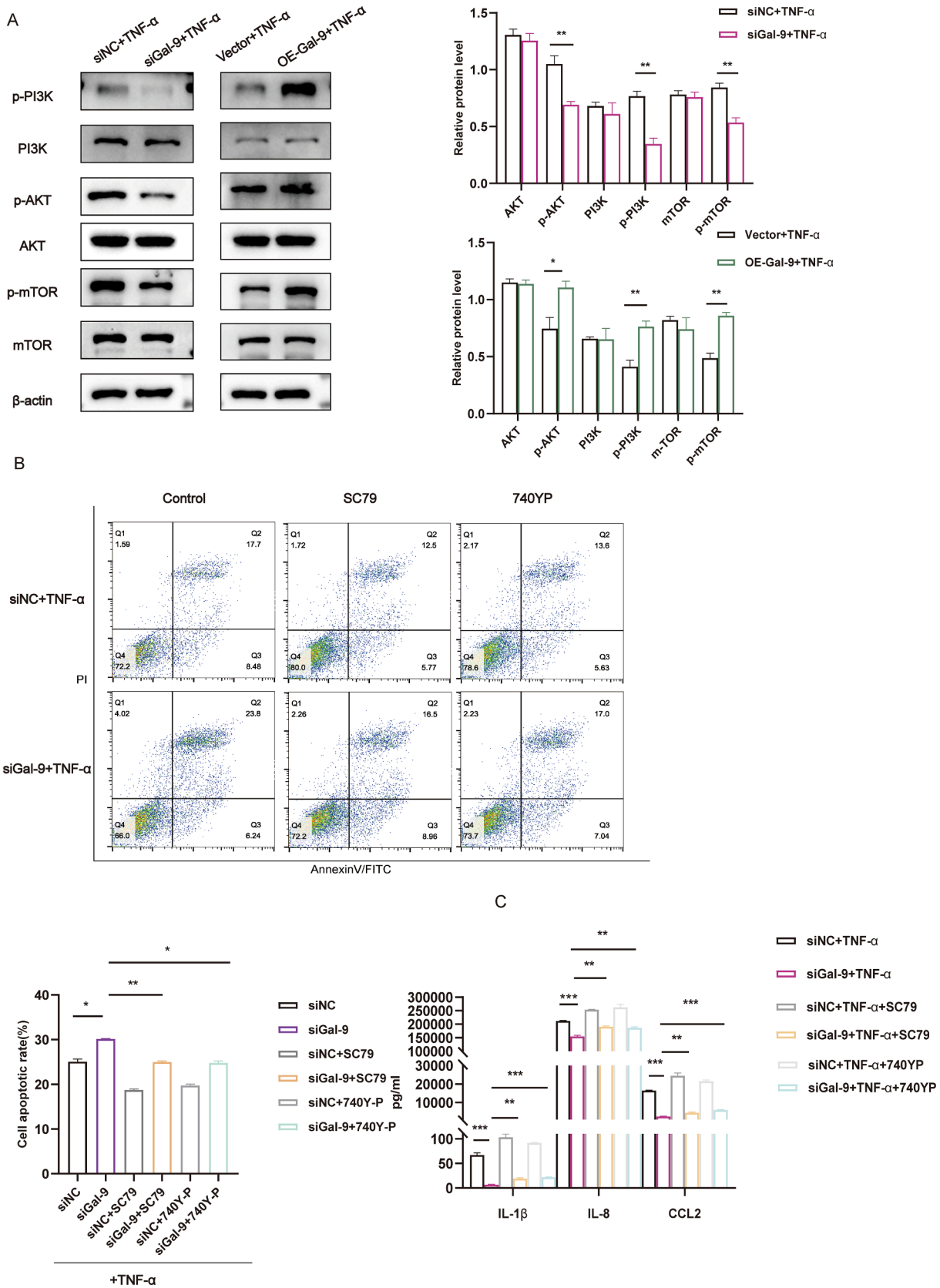


Fig. 6. Gal-9 knockdown promoted apoptosis and inhibited inflammatory factor through PI3K-AKT signaling pathway. Total and phosphorylated protein levels of PI3K/AKT/mTOR signaling pathway were detected through western blot after Gal-9 knockdown and overexpression (A). Apoptosis of RA-FLSs were detected via using flow cytometry (B) and inflammatory factor were measured using ELISA (C), when 740YP (PI3K agonist), SC79 (AKT agonist) were added. Quantitative results are shown as mean ± SEM, and the cell experiments were repeated at least three times. Statistical comparisons between two groups were made using Student's *t*-test. Multiple group comparisons were performed using one-way analysis of variance (ANOVA) followed by Dunnett's for multiple comparisons. $*p < 0.05$, $**p < 0.01$, $***p < 0.001$. Values of$p < 0.05$ were considered significant.

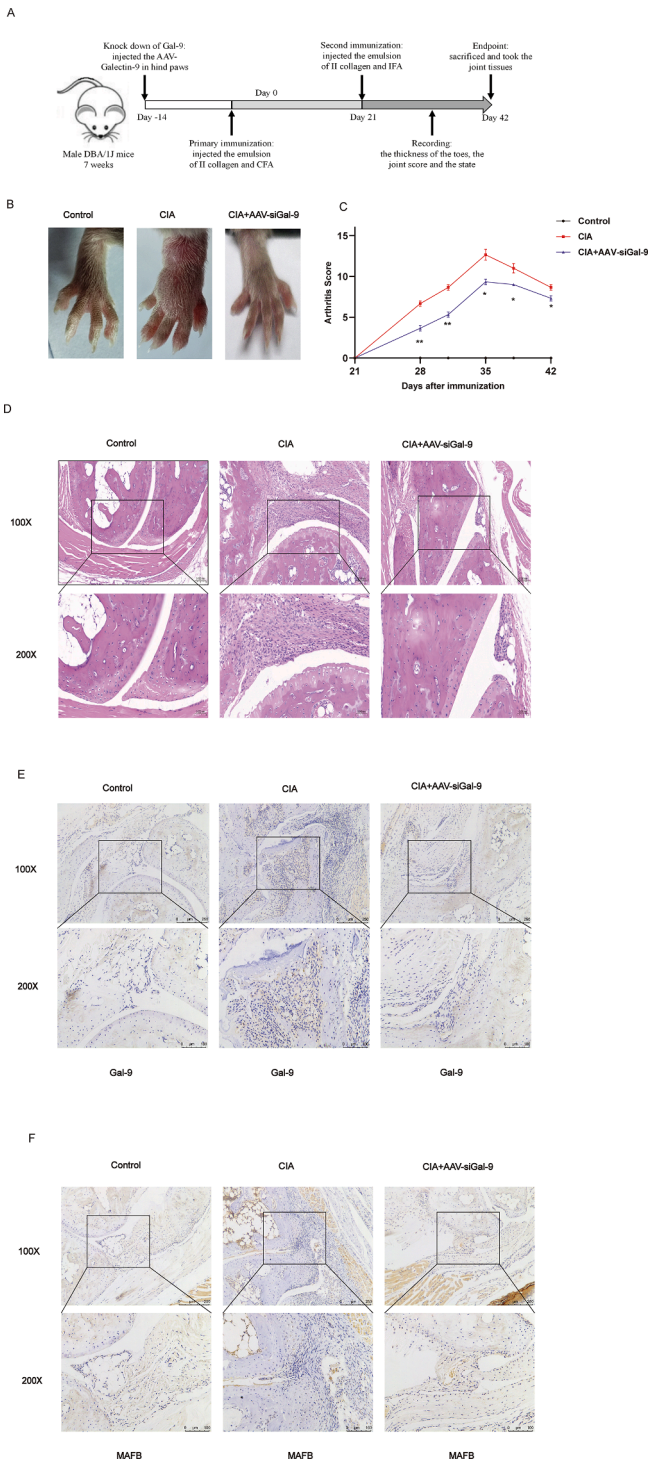


Fig. 7. Knockdown of Gal-9 prevent the CIA model development. The schematic of CIA mice model (A). The photographs of hind paw (B) and arthritis scores in control group, CIA group, CIA + AAV-siGal-9 group (C). Representative images of the synovium inflammation in each group with H&E staining (D). Gal-9 and MAFB expression was detected using IHC (E, F). Quantitative results are shown as mean ± SEM, and n = 4 per group. Multiple group comparisons were performed using one-way analysis of variance (ANOVA) followed by Dunnett's for multiple comparisons. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. Values of $p < 0.05$ were considered significant.

MAFB is a member of the MAF transcription factor family, which plays an important role in the development of various tissues. For example, MAFB was involved in the differentiation of macrophages and dendritic cells [50], acted synergistically with glucocorticoid receptors to induce tolerance in dendritic cells [51], activated insulin expression in β-cells, and was essential for maintaining β-cell function [52]. Multiple miRNAs modulated immune cell differentiation [53] inhibited osteosarcoma cell differentiation, regulated β-cell proliferation and apoptosis [54], and induced osteoclast differentiation [55]by targeting MAFB. In renal diseases, knockdown of MAFB in podocytes promoted the development of focal segmental glomerulosclerosi [56]. MAFB triggered the activation of M2 macrophages, thus supporting the repair of acute kidney injury [57]. Macrophages upregulated MAFB after acute kidney injury, which facilitated the expression of anti-inflammatory genes and contributed to the regression of inflammation [58]. Studies on MAFB in RA had focused on the polarization of macrophages; upregulation of MAFB promotes the activation of M2-type macrophages to exert anti-inflammatory effects [51,59,60]. Few studies had investigated whether MAFB plays a biological role in FLSs. RNA-seq transcriptome sequencing of the knockdown Gal-9 group suggested elevated MAFB mRNA expression, which was verified by protein level using western blot. Transfection of RA-FLSs using siRNA and overexpressed plasmids revealed that MAFB overexpression inhibited the proliferation of RA-FLSs, contributed to promote apoptosis, suppress migration and invasion, thereby attenuating activation of FLSs. In a study on post-injury neuroinflammation, activation of RARa receptors attenuated neuroinflammation by promoting M1-M2 phenotypic olarization of microglia and regulating the MAFB/Msr1/PI3K-AKT/NF-kB pathway [61]. In our study, interference with Gal-9 inhibited TNF-α induced activation of RA-FLSs via the MAFB and restored synovial homeostasis, providing a new option for the treatment of RA.

The PI3K-AKT pathway is involved in the abnormal biological behaviour and inflammatory responses in RA [62]. Semaphorin 5A enhances GPX4 expression and SREBP1/SCD- 1 signaling via the PI3K/AKT/mTOR signaling pathway, thereby suppressing ferroptosis of SFs [30]. PPAR-γ is a ligand-induced transcription factor critical for the inflammation, proliferation, and invasion of FLSs in RA [63]. In our study, knockdown of Gal-9 inhibited the PI3K-AKT-mTOR pathway to stimulate induced apoptosis and diminished the secretion of IL-1β, CCL2, MMP1, and MMP13. Rescue experiments further established that activation of PI3K-AKT pathway partially restrained biological behaviours induced by downregulation of Gal-9.

The current study has several limitations. First, we mainly investigated the role of endogenous Gal-9 on TNF-α-induced FLSs activation; however, it did not elaborate on whether Gal-9 would secondarily affect the biological behavior of other cells (e.g., T cells, macrophages). Second, the number of samples of clinical specimens from patients with RA is small. Finally, interfering with Gal-9 could inhibit activation of TNF-α-induced FLSs through the MAFB and PI3K-AKT pathways respectively, while the interaction between the MAFB and AKT pathways remains to be further verified.

In summary, our study showed that Gal-9 knockdown inhibited the proliferation, migration, invasion, expression of cytokines and MMPs, and facilitated apoptosis in TNF-α-induced RA-FLSs, via the MAFB and PI3K/AKT pathway, suggesting Gal-9 potential use as a therapeutic target.

Funding

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Ethical Statement

The study was conducted in accordance with the Declaration of Helsinki, and approved by the Linyi People's Hospital (Approval no. YX200237) and Qilu Hospital of Shandong University (Approval no. KYLL-202000-165-1). The animal study protocol was approved by the Shandong University (Approval no. 23025).

Informed consent was obtained from all subjects involved in the study. Written informed consent has been obtained from the patient(s) to publish this paper.

CRediT authorship contribution statement

Qian Jia: Conceptualization, Methodology, Validation, Visualization, Writing – original draft. **Qincheng Che:** Software, Validation, Visualization, Writing – original draft. **Xiaoyu Zhang:** Formal analysis. **Jie Chen:** Formal analysis. **Chunfeng Ren:** Investigation. **Yunpeng Wu:** Resources. **Weiqiang Liang:** Resources. **Xiaojie Zhang:** Formal analysis. **Yanshan Li:** Investigation. **Zunzhong Li:** Formal analysis. **Zhenchun Zhang:** Resources. **Qiang Shu:** Conceptualization, Methodology, Validation, Visualization, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors are unable or have chosen not to specify which data has been used.

Acknowledgments

Not applicable.

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